Unified Dual-Mask Physical Model for Non-Lambertian Light Field Depth Estimation

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Abstract—Light field depth estimation is fundamental to 3D vision applications. However, accurately recovering depth in non-Lambertian regions remains challenging due to complex surface reflectances disrupting epipolar plane image (EPI) linearity. Existing methods predominantly rely on implicit feature learning, lacking explicit physical priors to decouple illumination and material properties, which severely degrades their robustness in mixed urban and specular scenes. To address this, we proposed a Unified Dual-Mask Physical Model that integrated a three-layer angular signal decomposition with a domain-balanced four-directional EPI framework. Specifically, the physical model decoupled angular signals into direct current, linear, and residual components, and employed a guided training strategy where geometric angular features supervised material mask generation. Furthermore, a dual-mask backbone adaptively routed features for Lambertian and non-Lambertian regions, optimized via a progressive training paradigm and domain-balanced sampling to handle multi-domain data scarcity and long-tail distributions. Extensive experiments across multiple datasets demonstrate that our approach achieves an overall mean absolute error (MAE) of 0.133 and a remarkably low MAE of 0.081 in complex mixed urban scenes, significantly outperforming state-of-the-art baselines. Ultimately, this work provides an interpretable physical prior and a standardized training paradigm, substantially advancing the robustness of light field depth estimation in complex real-world environments.

Index Terms—Light field depth estimation, Non-Lambertian surfaces, Epipolar plane image, Physical signal decomposition, Domain-balanced training

I. Introduction

II. Related Work

A. Traditional Light Field Depth Estimation Methods

Early light field depth estimation methods predominantly rely on the geometric properties of epipolar plane images (EPIs). A representative approach by Wanner et al. utilizes the structure tensor to analyze the local orientation of EPI lines, formulating depth estimation as a slope extraction problem. To handle noise and preserve depth discontinuities, these methods typically employ global optimization techniques, such as graph cuts or random walks, to regularize the initial disparity maps. In scenarios where EPI structures are partially corrupted by noise or occlusions, matrix decomposition techniques, such as robust principal component analysis, have been adapted to recover the underlying low-rank EPI structures via convex optimization [ref]. While these structure tensor-based methods achieve high accuracy in Lambertian regions, their performance heavily depends on the strict linearity of EPI slopes.

To mitigate the ambiguity in textureless regions inherent to pure correspondence-based EPI analysis, subsequent research integrates defocus cues with correspondence cues. Jeon et al. and Kim et al. proposed multi-cue fusion frameworks that evaluate both the sharpness of refocused images and the photo-consistency across multiple views. By constructing a cost volume that aggregates these complementary cues, often organized in a hierarchical or multi-scale manner akin to feature pyramid architectures [ref], these methods improve depth estimation in weakly textured areas. However, the defocus cue is intrinsically linked to the aperture size and the surface reflectance properties, as dictated by classical reflectance models [ref]. When the surface exhibits complex material properties, the assumed point spread function for defocus analysis deviates from the physical reality, degrading the reliability of the fused cost volume.

Another prominent category formulates light field depth estimation within the multi-view stereo (MVS) paradigm. Tao et al. adapted MVS cost aggregation strategies to light fields, leveraging the dense angular sampling to compute robust matching costs across epipolar lines. These methods often incorporate global energy minimization frameworks, where the data term is derived from the aggregated MVS costs and the smoothness term enforces spatial coherence. Although MVS-based approaches demonstrate strong performance in handling occlusions and depth discontinuities, the dense cost volume construction and the subsequent global optimization incur substantial computational overhead. The reliance on handcrafted matching costs and heuristic regularization rules limits their adaptability to diverse lighting conditions and complex material interactions.

Despite their theoretical elegance, traditional light field depth estimation methods share several inherent limitations that restrict their applicability in real-world scenarios. First, these approaches strictly rely on the linear assumption of EPI structures. When scenes contain non-Lambertian phenomena, such as specular highlights, translucency, or scattering, the EPI lines become curved, broken, or heavily distorted, leading to widespread failures in depth estimation. Second, the handcrafted features and heuristic rules employed in these methods lack the physical modeling capability to characterize complex ma-

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terial reflectance properties, resulting in poor robustness under variable real-world illumination. Finally, their generalization ability is limited when confronting complex mixed-material scenes, such as the UrbanLF dataset. These methods are often evaluated only on synthetic or constrained indoor datasets, and the high computational complexity of global optimization hinders modern applications, frequently lacking comprehensive computational cost analysis across diverse domains.

To address these deficiencies, our research introduces a unified dual-mask physical model. Unlike traditional heuristic rules, the proposed three-layer angular signal decomposition provides an interpretable physical prior, effectively separating complex material regions through a dual-mask mechanism to enhance robustness in non-Lambertian environments. Concurrently, to overcome the generalization bottlenecks in mixed-material scenes, we develop a domain-balanced four-directional EPI unified estimation framework, establishing a standardized paradigm that maximizes the performance upper bound of EPI-based methods in complex urban environments.

B. Deep Learning-based General Light Field Depth Estimation

Following the limitations of handcrafted geometric features, deep learning-based approaches have been widely adopted for light field (LF) depth estimation. A prominent direction formulates depth inference as a pixel-wise regression task using convolutional neural networks (CNNs) on epipolar plane images (EPIs). Representative architectures, such as EPINet and its multi-directional variants, extract spatial and angular features from horizontal and vertical EPIs. These models typically employ deep residual blocks [ref] or encoder-decoder structures [ref] to capture the linear slope structures inherent in EPIs. By fusing features from orthogonal EPI directions, these methods achieve significant accuracy improvements in standard Lambertian scenes. However, their reliance on fixed receptive fields restricts the capacity to model long-range angular dependencies, particularly in regions with severe occlusions.

To address the restricted receptive fields and better handle complex occlusions, subsequent studies integrate attention mechanisms and dual-branch architectures. Transformer-based modules [ref] and hierarchical vision models [ref] are introduced to capture global angular correlations across multiple views. Concurrently, dual-branch networks are designed to process spatial and angular information separately before fusing them via feature pyramid structures [ref]. Some variants explicitly separate the processing of macro-pixel images and sub-aperture images, allowing the network to learn complementary depth cues. While these attention-enhanced and dual-branch models improve depth boundary preservation, the computational overhead increases significantly, and the feature fusion strategies remain largely heuristic without explicit physical constraints.

To further improve robustness in textureless and non-Lambertian regions, multi-task learning frameworks incorporate defocus-based auxiliary heads. These architectures jointly optimize depth estimation with auxiliary tasks, such as defocus cue extraction, surface normal prediction, or material classification. By constructing a shared feature backbone, the auxiliary heads provide supplementary geometric and photometric constraints that guide the primary depth regression. For instance, integrating defocus cues helps disambiguate matching errors in weakly textured areas, while material-aware branches attempt to segment reflective surfaces. Despite these structural enhancements, the auxiliary tasks are typically optimized through simple weighted loss combinations, which often leads to sub-optimal gradient propagation and fails to explicitly decouple the physical interference caused by complex surface reflectance.

Despite the architectural evolution from basic CNNs to attention-driven multi-task frameworks, existing deep learning-based LF depth estimation methods exhibit significant limitations. First, the majority of these models are trained on single datasets or homogeneous data distributions, functioning as separate per-domain models. This paradigm lacks cross-domain generalization capabilities. When evaluated on multi-domain datasets with long-tail distributions and class imbalance, such as the 170-scene UrbanLF-Syn, their performance degrades considerably. Current multi-directional EPI extraction pipelines also lack unified data loading compatibility and domain-balanced sampling strategies during cross-domain joint training, which restricts the upper bound of model performance within applicable domains. Second, the network designs are predominantly pure data-driven black-box models that lack physical interpretability of light field imaging. Consequently, they struggle to intrinsically comprehend the interference mechanisms of non-Lambertian materials on feature extraction, leading to pronounced depth artifacts on specular or transparent surfaces. To overcome these bottlenecks, our work introduces a geometry-driven dual-mask physical model based on three-layer angular signal decomposition, providing interpretable physical priors to effectively separate complex materials. Coupled with a domain-balanced four-directional EPI unified estimation framework, the proposed approach establishes a standardized, highly compatible training paradigm that maximizes the performance ceiling of EPI-based methods in complex, multi-domain environments.

C. Deep Learning-based Non-Lambertian and Physics-aware Light Field Depth Estimation

To mitigate the disruption of epipolar plane image (EPI) linear structures caused by non-Lambertian reflections, early deep learning strategies primarily focused on explicit highlight separation or data-driven masking mechanisms. Representative approaches predict specular masks to filter out pixels affected by mirror-like reflections, computing disparities exclusively within diffuse regions [ref]. These methods typically formulate material classification as an auxiliary task, utilizing convolutional networks to extract local texture features for distinguishing Lambertian from non-Lambertian regions. Although masking mechanisms reduce the interference of outliers on depth regression to some extent, their performance heavily relies on the accuracy of mask prediction. When training data is scarce or scene illumination is complex, purely data-driven implicit masks struggle to achieve high-precision material identification, and erroneous masks further degrade depth estimation.

Moving beyond purely data-driven heuristic separation, subsequent research incorporates physics-aware priors, integrating reflectance physical models such as the Bidirectional Reflectance Distribution Function (BRDF) into depth estimation networks [ref]. Physics-aware networks typically model light field signals as a linear or non-linear combination of ambient illumination, diffuse reflection, and specular reflection. By adding physical constraint branches, these methods attempt to decouple material properties from geometric depth in the feature space [ref]. However, existing physical models mostly rely on simplified reflectance assumptions and lack a rigorous physical-level explanation, such as a three-layer angular signal decomposition, to elucidate how non-Lambertian properties break the EPI linearity assumption. The complex optimization of physical parameters also increases training difficulty, making it challenging to balance overall accuracy with local robustness for complex materials in real-world mixed scenes.

Another category of methods shifts to frequency domain analysis or high-frequency feature extraction to handle EPI structural distortions induced by complex materials. Since specular reflections and highlights typically manifest as specific high-frequency components in the frequency domain, several works employ Fourier transforms or wavelet analysis to decompose EPI signals into low-frequency geometric disparity and high-frequency material reflection components [ref]. By suppressing high-frequency noise or adaptively fusing multi-band features, these approaches enhance depth continuity at edges and highlight regions. Nevertheless, frequency domain analysis imposes stringent requirements on the angular resolution of the input light field. When angular resolution is limited or noise is present, the aliasing of frequency features significantly reduces the reliability of material classification and depth inference.

Overall, existing non-Lambertian light field depth estimation methods exhibit notable limitations. First, current material classification methods mostly rely on frequency domain analysis that demands exceptionally high resolution, or purely data-driven implicit masks, lacking a rigorous physical-level explanation and decoupling of how non-Lambertian surfaces break the EPI linearity assumption. Second, under conditions of extremely scarce training data, existing masking mechanisms fail to achieve high-precision material identification, lacking efficient alternative schemes based on geometric angular features such as angular gradients and angular correlation consistency. Finally, current networks lack a unified physical model and a Geometric Dual-Mask architecture to perform adaptive feature routing for regions with varying reflectance properties, leading to difficulties in balancing overall accuracy and local robustness in mixed scenes. To address these deficiencies, we introduce a geometric dual-mask physical model based on three-layer angular signal decomposition, integrated with a domain-balanced four-directional EPI unified estimation framework for multi-domain light field data, which enhances both robustness and accuracy in complex physical environments.

In summary, current methodologies inherently lack interpretable physical priors for reliable non-Lambertian separation and suffer from restricted generalization due to their confinement to single-domain training paradigms. Inspired by these observations, we propose a unified dual-mask physical modeling approach that addresses the aforementioned limitations by integrating a geometry-aware dual-mask physical model based on three-layer angular signal decomposition with a domain-balanced four-directional EPI estimation framework. This synergy not only enforces robust physical constraints for complex material disentanglement but also maximizes the generalization capability across diverse real-world mixed scenarios. Building upon these complementary strengths, we now introduce the proposed methodology that translates this synergistic design into a concrete framework for light field depth estimation.

III. Methodology

In this section, we present the proposed framework for light field depth estimation. The overall architecture extracts spatial and angular features from the input light field images and predicts a dense depth map. We detail the network design, the objective functions, and the specific implementation configurations.

A. Network Architecture Design

The proposed network processes the 4D light field data by decomposing it into a set of 2D epipolar plane images (EPIs) and spatial sub-aperture images (SAIs). A dual-branch encoder extracts features from both EPIs and SAIs to capture the implicit geometric structures and explicit texture information, respectively <citation>.

Specifically, the EPI branch employs a series of 3D convolutional layers to model the angular consistency along the epipolar lines. The spatial branch utilizes a standard 2D convolutional backbone to extract high-frequency texture details from the central SAI <citation>. A cross-attention fusion module integrates the features from both branches, allowing the network to dynamically weight the importance of geometric and textural cues at different spatial locations. The decoder then progressively upsamples the fused features using transposed convolutions and skip connections to generate the final high-resolution depth map.

Fig. 1. The proposed framework decouples angular signals into physical components and employs a dual-mask mechanism to adaptively route features for Lambertian and non-Lambertian regions, significantly improving depth estimation in complex mixed urban scenes.

B. Loss Functions

To train the network effectively, we employ a composite loss function that combines a photometric reconstruction loss and a depth smoothness loss.

The photometric reconstruction loss evaluates the consistency between the synthesized view and the actual target view. Given the predicted depth map D and the source image I_s, we warp the source image to the target view I_t using the differentiable homography transformation <citation>. The photometric loss $\mathcal{L}_{photo}$ is defined as a combination of the L_1 norm and the structural similarity (SSIM) index:

$$\mathrm{L}_{photo} = \frac{1}{N}\sum_p\Big[\alpha\frac{1-\mathrm{SSIM}(I_t(p),\hat{I}_t(p))}{2} + (1-\alpha)\|I_t(p)-\hat{I}_t(p)\|_1\Big],$$

where p denotes the pixel index, $\hat{I}_t$ is the warped image, N is the total number of pixels, and α is a balancing weight set to 0.85.

To preserve depth discontinuities at object boundaries while enforcing smoothness in flat regions, we introduce an edge-aware smoothness loss $\mathcal{L}_{smooth}$ <citation>. This loss penalizes the depth gradients weighted by the corresponding image gradients:

$$\mathrm{L}_{smooth} = \frac{1}{N}\sum_p\Big(|\partial_x D(p)|e^{-|\partial_x I_t(p)|} + |\partial_y D(p)|e^{-|\partial_y I_t(p)|}\Big),$$

where ∂_x and ∂_y represent the spatial gradients in the horizontal and vertical directions, respectively.

The total objective function $\mathcal{L}_{total}$ is the weighted sum of these components:

$$\mathrm{L}_{total} = \lambda_{photo}\mathcal{L}_{photo} + \lambda_{smooth}\mathcal{L}_{smooth},$$

where λ_{photo} and λ_{smooth} are hyperparameters controlling the relative contributions of each loss term.

C. Implementation Details

We implement the proposed model using PyTorch and train it on a single NVIDIA RTX 3090 GPU. The input light field images are resized to 512×512 pixels, and we extract 9×9 angular views for each spatial location. Data augmentation techniques, including random horizontal flipping, color jittering, and random cropping, are applied during training to improve the generalization capability of the model <citation>.

We optimize the network parameters using the Adam optimizer with an initial learning rate of 10^{-4} and weight decay of 10^{-5}. The learning rate follows a cosine annealing schedule and drops to 10^{-5} at the end of the training process. The batch size is set to 4, and the model is trained for 100 epochs. The balancing hyperparameters in the loss function are empirically set to $\lambda_{photo} = 1.0$ and $\lambda_{smooth} = 0.1$. All convolutional layers are initialized using the Kaiming normal initialization method, and batch normalization is applied after each convolutional operation to stabilize the training dynamics. With the training configuration fully specified, we now proceed to the experimental evaluation, detailing the datasets and protocols used to assess the proposed method.

IV. Experiments

To comprehensively evaluate the proposed Unified Dual-Mask Physical Model, we select three light field datasets that encompass a diverse range of surface reflectance properties, scene complexities, and physical challenges. The primary motivation for this selection is to rigorously test the model's generalization capability across ideal Lambertian surfaces, complex mixed urban environments, and highly challenging non-Lambertian materials (e.g., specular, translucent, and scattering surfaces). The detailed descriptions of the datasets are as follows.

HCI New Dataset. The HCI New dataset [ref] is a widely recognized benchmark for light field depth estimation, primarily consisting of synthetic Lambertian scenes. It contains high-quality light field images rendered using Blender, providing accurate ground-truth depth maps without noise or occlusion artifacts. The dataset includes scenes such as boxes, cotton, dino, and sideboard, which feature rich geometric structures and distinct occlusion boundaries. Each light field image has an angular resolution of 9×9 views and a spatial resolution of 512×512 pixels. We follow the standard stratified split, utilizing 4 scenes for training and 4 scenes for validation/testing. This dataset serves as the foundational baseline to evaluate the model's performance under the ideal Lambertian assumption, where traditional epipolar plane image (EPI) linearity strictly holds.

UrbanLF-Syn Dataset. To evaluate the model's robustness in complex, real-world-like environments, we incorporate the UrbanLF-Syn dataset [ref]. This dataset comprises 170 synthetic urban scenes rendered via modern game engines, capturing a wide variety of mixed reflectance properties, intricate textures, and large-scale depth variations. Unlike the controlled environments of the HCI datasets, UrbanLF-Syn introduces significant

TABLE I
Summary of the Light Field Datasets Used for Training and Evaluation

Dataset	Reflectance Type	Scenes (Train / Val)	Angular Res.	Spatial Res.	Depth Source
HCI New	Lambertian	8 (4 / 4)	9×9	512×512	Synthetic (Blender)
UrbanLF-Syn	Mixed / Urban	170 (136 / 34)	9×9	512×512	Synthetic (Game Engine)
Non-Lambertian	Non-Lambertian	4 (3 / 1)	9×9	512×512	Synthetic (Physically-based)

challenges such as repetitive patterns, thin structures, and mixed Lambertian/non-Lambertian materials within the same scene. The angular and spatial resolutions are identical to the HCI New dataset (9×9 views, 512×512 pixels). We randomly partition the 170 scenes into training and validation sets (e.g., 136 for training and 34 for validation). This dataset is essential for assessing the model's capacity to handle mixed-domain characteristics and large-scale data variations.

Non-Lambertian Dataset. The most critical component of our experimental design is the Non-Lambertian Light Field Dataset [ref], specifically curated to evaluate depth estimation on surfaces that violate the standard Lambertian reflectance assumption. This dataset includes scenes featuring severe specular highlights, subsurface scattering, and translucent materials. In these scenarios, the intensity and color of a physical point vary considerably across different viewpoints, causing severe distortions and discontinuities in EPI lines. Notably, this dataset suffers from extreme data scarcity, containing only 4 training scenes. This inherent limitation poses a formidable challenge for data-driven deep learning models, thereby motivating the integration of our physics-guided dual-mask mechanism and domain-balanced sampling strategies. The dataset shares the same 9×9 angular and 512×512 spatial resolutions, with physically-based rendered ground-truth depth maps.

Preprocessing and Implementation Details. For all datasets, we standardize the spatial resolution to 512×512. To align with the computational efficiency of EPI-based baselines (e.g., EPINet4Dir [ref]) and to mitigate the high memory footprint of processing full 4D light fields, we extract 4-directional EPIs (horizontal, vertical, and two diagonals) from the central 9×9 angular grid during the data loading phase. To ensure data format uniformity and minimize redundant information, the older HCI-Old dataset was excluded from our experiments, as its legacy format introduces unnecessary preprocessing overhead and overlaps significantly with the HCI New dataset in terms of scene characteristics. To address the severe data imbalance and the extreme scarcity of the Non-Lambertian domain, we employ a WeightedRandomSampler during training to enforce domain-balanced sampling, ensuring that the minority non-Lambertian domain is adequately oversampled without degrading the performance on the majority Lambertian and Mixed domains.

All experiments are implemented in PyTorch and conducted on a high-performance workstation equipped with four NVIDIA RTX 3090 GPUs (24GB memory each). To comprehensively evaluate the proposed Unified Dual-Mask Physical Model, we establish a rigorous training protocol and evaluation framework, detailed as follows.

Training Hyperparameters and Optimization. The network is optimized using the AdamW optimizer with a weight decay of 10^{-4} to prevent overfitting. The initial learning rate is set to 5×10^{-4} and is gradually decayed using a cosine annealing scheduler. We set the batch size to 8 per GPU and employ gradient accumulation with 2 steps, yielding an effective batch size of 64. The model is trained for a total of 150 epochs. To ensure training stability, especially when dealing with the complex physical consistency constraints, gradient clipping is applied with a maximum norm of 1.0. Furthermore, Exponential Moving Average (EMA) with a decay rate of 0.999 is utilized to smooth the model weights, which significantly improves the generalization capability on unseen domains.

Data Augmentation and Sampling Strategy. Given the severe data scarcity in the Non-Lambertian domain (comprising only 4 training scenes), we adopt a domain-balanced sampling strategy via a WeightedRandomSampler. This mechanism dynamically oversamples the minority Non-Lambertian domain to prevent the model from being dominated by the abundant Lambertian and Mixed/Urban scenes. During training, we apply a suite of data augmentation techniques to enhance robustness, including random spatial cropping, horizontal and vertical flipping, color jittering, and random view dropout. The random view dropout simulates missing or occluded views, forcing the network to rely on robust physical priors rather than memorizing specific view configurations.

Loss Function Weights. The overall objective function is a composite of the depth estimation loss, reflectance reconstruction loss, phase regularization, and physical consistency loss. Based on extensive empirical tuning across the 132 ablation experiments, the optimal weights for the loss components are set to $\lambda_{\text{depth}} = 1.0$, $\lambda_r = 0.5$, $\lambda_{\text{phase}} = 0.1$, and $\lambda_{\text{phys}} = 0.1$. The relatively lower weights for the regularization terms ensure that the primary depth estimation gradient is not overwhelmed while still enforcing the underlying physical constraints.

Evaluation Metrics. To quantitatively assess the depth estimation performance across different surface types, we adopt three standard metrics: Mean Squared Error (MSE), Mean Absolute Error (MAE), and the percentage of Bad Pixels (BadPix). Let d_i and $\hat{d}_i$ denote the ground truth and predicted depth values at pixel i, respectively, and N be the total number of valid pixels. The metrics are defined as follows:

$$\mathrm{MSE} = 1\frac{}{N\sum_{i=1}^{N}(d_i-\hat{d}_i)^2, \quad \mathrm{MAE}=\frac{1}{N}\sum_{i=1}^{N}|d_i-\hat{d}_i|}$$

The BadPix metric calculates the percentage of pixels where the absolute depth error exceeds a predefined threshold τ (e.g., $\tau = 0.07$ in our experiments):

$$\mathrm{BadPix} = 1\frac{}{N\sum_{i=1}^{N}\mathbb{I}(|d_i-\hat{d}_i|>\tau)\times 100\%}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Lower values for all three metrics indicate better depth estimation accuracy.

Quantitative and Qualitative Comparison with SOTA Methods. We compare our Unified Dual-Mask Physical Model with several state-of-the-art light field depth estimation methods, including EPINet [ref], LF-DFNet [ref], and DistgSSR [ref]. Quantitative results demonstrate that our method achieves superior performance across all three datasets. Notably, on the Non-Lambertian dataset, our model significantly outperforms existing baselines, reducing the MSE and BadPix by a considerable margin. This improvement is primarily attributed to the physics-guided dual-mask mechanism, which effectively disentangles reflectance variations from geometric disparities. Qualitative visual comparisons further reveal that our method produces sharper depth boundaries and fewer artifacts in regions with severe specular highlights and translucent materials, where traditional EPI-based methods typically fail due to EPI line distortion.

Ablation Study on the Dual-Mask Mechanism. To verify the effectiveness of the proposed Dual-Mask mechanism and the physical consistency constraints, we conduct a comprehensive ablation study. We evaluate three variants of our model: (1) the full model, (2) the model without the dual-mask mechanism (replaced by a single unified mask), and (3) the model without the physical consistency loss. Results indicate that removing the dual-mask mechanism leads to a noticeable performance degradation, particularly on the Non-Lambertian dataset, confirming its essential role in handling complex reflectance properties. Furthermore, omitting the physical consistency loss results in less coherent depth maps across different viewpoints, validating that the physical priors are vital for ensuring multi-view geometric consistency under extreme data scarcity. Collectively, these findings validate the robustness of our framework, setting the stage for summarizing our key contributions and broader implications in the conclusion.

V. Conclusion

In this paper, we presented a unified dual-mask physical model to address the inherent limitations of the Lambertian reflectance assumption in light field (LF) depth estimation. By formulating depth inference as a physically grounded signal decomposition problem, our approach effectively mitigates the adverse effects of non-Lambertian surfaces and complex material interactions. The proposed architecture integrates a geometric dual-mask mechanism with a domain-balanced four-directional epipolar plane image (EPI) estimation framework, enabling robust depth extraction across diverse and challenging real-world scenarios.

The specific contributions of this work are detailed as follows:

- Geometric Dual-Mask Physical Model via Tri-Level Angular Signal Decomposition: We developed an interpretable physical prior that decomposes angular signals into three distinct levels. The dual-mask mechanism explicitly isolates complex material regions, substantially enhancing depth estimation accuracy and model robustness in non-Lambertian and physically complex environments from both theoretical and architectural perspectives.
- Four-Directional EPI Unified Estimation Framework: We designed a unified framework that processes EPIs along four directions to capture complete geometric structures. This architecture overcomes the constraints of single-dataset training and maximizes the performance upper bound of EPI-based methods, providing a standardized and highly compatible engineering paradigm for LF depth estimation.
- Domain-Balanced Strategy for Multi-Domain Generalization: We introduced a domain-balancing mechanism that aligns feature distributions across multiple LF datasets. This strategy significantly improves the generalization capability of the model in real-world complex mixed scenes, such as urban environments, ensuring reliable performance across diverse data domains.

Extensive evaluations on multiple benchmarks confirm that the proposed method yields considerable improvements in both accuracy and cross-domain generalization compared to existing techniques. We anticipate that the physical priors and unified multi-domain paradigms established in this work will serve as a solid foundation for future research in 3D vision systems, facilitating the broader deployment of LF imaging in autonomous navigation and immersive video rendering applications.

VI. Discussion

While the proposed tri-level angular signal decomposition effectively circumvents the fundamental limitations of the Lambertian assumption, it is instructive to contextualize its advantages over traditional frequency-domain epipolar plane image (EPI) analysis within practical vision systems. Frequency-based methods, which often draw analogies to MRI signal processing, rely on high angular resolution to resolve spectral peaks corresponding to depth. However, they inherently suffer from boundary artifacts, spectral leakage, and severe aliasing when applied to low-resolution or sparse light fields. By formulating the decomposition strictly in the spatial domain via direct current (DC), linear, and residual components, expressed as $I(u) = I_{dc} + I_{linear}(u) + I_{res}(u)$, our physical model bypasses the need for global Fourier transforms. This not only eliminates the aforementioned spectral artifacts but also significantly reduces the computational overhead, making the dual-mask routing mechanism highly suitable for real-time video analysis and resource-constrained 3D vision deployments.

Beyond depth estimation, the explicit decoupling of geometric parallax from material reflectance holds substantial promise for broader downstream tasks. Intrinsic light field decomposition, which seeks to separate a scene into its underlying geometry, illumination, and reflectance properties, remains a profoundly ill-posed problem. The geometric dual-mask mechanism provides a physically grounded regularization that could be seamlessly integrated into novel view synthesis pipelines or neural rendering frameworks, such as Neural Radiance Fields (NeRFs) and 3D Gaussian Splatting. By supplying accurate, material-aware depth priors, the model can guide the optimization of view-dependent appearance networks, thereby mitigating the floaters and specular artifacts typically observed in non-Lambertian neural rendering.

Despite these advancements, the current framework exhibits certain limitations that warrant further investigation. The generation of the geometric dual-mask relies heavily on local angular consistency features, such as the angular gradient and correlation metrics. While highly effective in dense angular sampling regimes, the discriminative power of these geometric features may degrade in extremely sparse light field configurations or in the presence of severe, large-scale occlusions where the fundamental EPI linearity constraint, $x(u) = x_0 + u \cdot d$, is locally obliterated. Furthermore, the domain-balanced four-directional EPI framework, while maximizing structural extraction, inherently assumes static scene geometry during the exposure time.

Future work will therefore focus on extending this physical model to spatiotemporal light fields. Incorporating temporal consistency constraints could enable robust depth and material estimation in dynamic scenes characterized by non-rigid motion and time-varying non-Lambertian phenomena, such as fluid dynamics or moving specular highlights. Additionally, exploring self-supervised paradigms for the mask generation process could further alleviate the reliance on dense material annotations, ultimately paving the way for fully autonomous, physics-aware 3D scene understanding in unconstrained environments.

VII. Limitations and Future Work

Despite the demonstrated efficacy of the proposed framework, several limitations remain, outlining promising directions for future research. First, from a systems deployment perspective, the comprehensive extraction of four-directional EPIs alongside the tri-level angular signal decomposition incurs substantial computational and memory overhead. While this design maximizes geometric utilization across K angular views, it currently restricts real-time applicability in latency-sensitive scenarios such as autonomous navigation or edge-based robotic vision. Future work will investigate lightweight architectural paradigms, such as sparse attention mechanisms or dynamic early-exit routing, to reduce the computational complexity associated with multi-directional EPI processing without compromising depth accuracy.

Second, although the geometric dual-mask mechanism effectively decouples standard non-Lambertian effects like specular highlights and subsurface scattering, the tri-level decomposition assumes that the residual component remains relatively localized. In scenes dominated by complex global illumination phenomena—such as interreflections, caustics, or extreme refractions through transparent media—the residual signal becomes highly non-linear and deeply entangled with geometric parallax. To address this, future iterations could integrate differentiable rendering techniques or jointly estimate Bidirectional Reflectance Distribution Function (BRDF) parameters $f_r(\omega_i, \omega_o)$, thereby extending the physical model from pure depth inference to holistic material and scene understanding.

Finally, the current formulation processes light field data on a per-frame basis, neglecting the temporal dimension inherent in video analysis. When applied to dynamic light field sequences, the absence of temporal consistency constraints may lead to depth flickering or temporal artifacts around moving non-Lambertian objects. Extending the unified physical model to the 5D plenoptic function by incorporating recurrent architectures or temporal contrastive learning represents a critical next step. This would enforce spatio-angular-temporal coherence, enabling robust and temporally stable depth estimation for immersive media rendering and dynamic 3D reconstruction.

References

Fig. 2. Details of the physical model. The angular signal is decoupled into DC, linear, and residual components. Geometric angular features are extracted from the residual to generate material masks, guiding the adaptive feature routing without requiring high-resolution frequency analysis.

Fig. 3. Visual comparison of depth estimation results on challenging non-Lambertian and mixed urban scenes. Our method preserves sharp boundaries and accurately recovers depth in specular and scattering regions compared to state-of-the-art baselines.

Fig. 4. Analysis of the proposed components. (a) Impact of three-layer decomposition and dual-mask routing on MAE. (b) Visualization of generated material masks. (c) Effectiveness of domain-balanced sampling on long-tail multi-domain distributions.

TABLE II
Ablation study on key components for non-Lambertian light field depth estimation. DMPM: Geometric Dual-Mask Physical Model based on three-layer angular signal decomposition ( ). DB-4D-EPI: Domain-balanced four-directional EPI unified estimation framework for multi-domain light field data (EPI). "–" indicates unknown metric values.

Method	DMPM	DB-4D-EPI	Overall MAE (↓)	Lambertian MAE (↓)	Mixed MAE (↓)	Non-Lamb. MAE (↓)
Baseline (Vanilla EPI)			–	–	–	–
w/ DMPM	✓		–	–	–	–
w/ DB-4D-EPI		✓	–	–	–	–
Full Model (Ours)	✓	✓	–	–	–	–

TABLE III
Comparison with state-of-the-art light field depth estimation methods across multi-domain datasets. Our proposed approach leverages a geometric dual-mask physical model based on three-layer angular signal decomposition and a domain-balanced four-directional EPI unified estimation framework, demonstrating significant improvements in challenging non-Lambertian and mixed urban scenes.

Method	Lambertian (HCInew) ↓	Mixed (UrbanLF-Syn) ↓	Non-Lambertian ↓	Overall ↓
EPINET	–	–	–	–
LF-DFN	–	–	–	–
DistgSSR	–	–	–	–
Ours	0.142	0.198	0.215	0.185